

Project JARVIS: Open Quadruped Robotics Platform (OQRP)

Based on the Robotics Engineering Development Framework (REDF v1.1)

The Golden Rule of Progression

For every version checkpoint (v0.1 to v1.0), you must complete this exact sequence before moving on. Never skip a step:

1. **Code:** Write and test the firmware.
2. **Video:** Record physical proof of the robot completing the milestone.
3. **Documentation:** Write the Project Report (Objectives, Components, Problems, Solutions, Photos, Lessons Learned).
4. **GitHub Push:** Commit the code, push the documentation, and tag the release (e.g., v0.1).
5. **Reflection:** Review what failed and how it influences the next version.

PHASE 1: Foundations (4–5 Weeks)

1. Mathematics & Dimensional Analysis

- **Angles & Trigonometry:** Degrees, Radians, Sine, Cosine, Tangent.
 - **Primary:** 3Blue1Brown — Essence of Trigonometry
 - **Secondary:** Khan Academy Trigonometry
 - **Practice:** Paul's Online Notes, Khan Academy Exercises
 - *Q1: Convert 45° to radians.*
 - *Q2: If a 10 cm leg link rotates 60° from the horizontal axis, how high is the tip?*
- **Coordinate Systems, Vectors & Matrices:** (X, Y, Z) , Vector Magnitude, Matrix Multiplication, Transformations.
 - **Primary:** 3Blue1Brown — Essence of Linear Algebra
 - **Secondary:** Khan Academy Linear Algebra
 - **Practice:** MIT OCW Linear Algebra Problems
 - *Q1: How do you multiply a 2×2 rotation matrix by a 2×1 position vector?*
 - *Q2: If the hip is at $(0,0,0)$ and the foot is at $(0, -5, -10)$, what is the magnitude of the vector connecting them?*

2. Physics & Mechanics

- **Force, Torque & Inertia:** Newton's Laws, Rotational Motion, Momentum.
 - **Primary:** Khan Academy Physics

- **Secondary:** Physics Classroom
- **Practice:** Solve numerical problems by hand
- *Q1: Long arm vs. short arm: Which requires more torque to lift the same payload?*
- **Stability & Center of Mass:** CoM, Support Polygon, Friction.
 - **Primary:** FRC (FIRST Robotics) Engineering Resources
 - **Secondary:** YouTube: James Bruton Robotics
 - *Q1: Where must the CoM remain vertically for the robot not to tip over?*

3. Basic Digital & Power Electronics

- **Digital Electronics:** HIGH/LOW, Pull-up, Pull-down, Floating Inputs.
 - **Primary:** Ben Eater Digital Electronics
 - **Secondary:** All About Circuits
 - *Q1: What happens if an input pin reading a limit switch is left floating?*
- **Voltage, Current & Power:** Ohm's Law, Power Law, Current Budgeting.
 - **Primary:** Paul McWhorter Electronics
 - **Secondary:** All About Circuits
 - *Q1: If a circuit runs at 5 V and draws 3 A, how many Watts of power is it consuming?*
- **Batteries & Regulators:** LiPo, Buck Converter, Power Distribution.
 - **Primary:** DroneBot Workshop
 - **Secondary:** GreatScott!
 - *Q1: Why must you NEVER power 9 servos directly from the ESP32?*
- **Reading Datasheets:** Pinouts, Timing Diagrams, Absolute Maximum Ratings.
 - **Primary:** Actual Component Datasheets
 - **Secondary:** DigiKey Tutorials
 - *Q1: Where in the ESP32 datasheet do you find the maximum allowable current per GPIO pin?*

4. CAD & Manufacturing

- **Fusion 360:** Sketches, Constraints, Assemblies, Joints.
 - **Primary:** Product Design Online (Fusion 360)
 - **Secondary:** Autodesk Learn

- *Q1: Why use parametric constraints instead of freehand sketches for a replacement leg?*
- **Manufacturing:** Fasteners, Tolerances, 3D Printing Basics.
 - **Primary:** Teaching Tech
 - **Secondary:** CNC Kitchen
 - *Q1: Why is thread locker mandatory on moving joint screws?*

5. Version Control & Linux Workspace

- **Linux & Ubuntu:** Terminal, File System, Permissions, SSH.
 - **Primary:** NetworkChuck Linux Course
 - **Secondary:** Linux Journey
 - **Practice:** Daily terminal usage
 - *Q1: What is the exact mv command to move a file from /downloads to your current directory?*
- **Git & GitHub:** Commit, Branch, Merge, Tags, Releases.
 - **Primary:** Git & GitHub Crash Course (Traversy Media)
 - **Secondary:** Git Documentation
 - *Q1: Why should every major physical milestone have a Git Release Tag?*

v0.1 Assembly Release & Verification

Safety & Functional Checklist:

- Fuse or current protection verified.
- Battery polarity verified before connection.
- Emergency power disconnects available.
- Chassis fully assembled; logic isolated from servo power.

Deliverables: v0.1 Git Tag, wiring diagram pushed, Project Report published.

PHASE 2: Programming & Embedded Systems (3–4 Weeks)

6. Core C++ & Algorithms

- **C++ Fundamentals:** Functions, Classes, Structs, Pointers, OOP.
 - **Primary:** The Cherno C++
 - **Secondary:** learncpp.com
 - **Practice:** LeetCode Easy Problems
 - *Q1: Why pass a large array of servo configurations by reference (pointer) rather than by value?*
- **Data Structures & Algorithms (DSA):** Arrays, Linked Lists, Stacks, Queues, Trees.
 - **Primary:** Striver A2Z DSA Sheet
 - **Secondary:** NeetCode

7. Embedded Software Reliability (NASA Power of 10)

- **Aerospace Rules:** No dynamic memory allocation, small functions, static analysis mindset.
 - **Primary:** NASA Power of 10 Paper
 - **Secondary:** Embedded Artistry
 - *Q1: Why is dynamic memory allocation (malloc/new) strictly forbidden in mission-critical embedded systems?*

8. PlatformIO & Embedded Control

- **PlatformIO:** Project Structure, Libraries, Build System.
 - **Primary:** Official PlatformIO Documentation
 - **Secondary:** PlatformIO YouTube Tutorials
- **ESP32 Core Control:** GPIO, PWM, Timers, Interrupts.
 - **Primary:** Random Nerd Tutorials
 - **Secondary:** ESP32 Documentation
 - *Q1: How do you use hardware timers to maintain a strict 50 Hz control loop without blocking delay()?*
- **Protocols (I2C & UART):** Communication protocols.
 - **Primary:** DroneBot Workshop
 - **Secondary:** DigiKey Tutorials
 - *Q1: How does the ESP32 independently control 16 servos using only two SDA/SCL lines?*

v0.2 Servo Control Release & Verification

Functional Tests:

- Individual servos reach commanded angles cleanly.
- No ESP32 brownouts under load.

Performance Metrics:

- Loop execution time (must maintain 50 Hz reliably).

Deliverables: v0.2 Git Tag, calibration offsets saved, Project Report pushed.

PHASE 3: Robotics Core (4–5 Weeks)

9. Servo Calibration & Mechanics

- **Calibration:** Mechanical zero, Offsets, Gear backlash, Deadband.
 - **Primary:** Servo Datasheet & DroneBot Workshop
 - *Q1: How do you program a "soft limit" to prevent stripping metal gears?*

10. Kinematics & Gait Generation

- **Forward Kinematics:** Coordinate Frames, Link Transformations.
 - **Primary:** Modern Robotics (Kevin Lynch)
 - **Secondary:** Robotics Back-End
 - *Q1: Given hip and knee angles, what is the trigonometric formula for the foot's position?*
- **Gait Generation:** Tripod/Trot gaits, Stability.
 - **Primary:** James Bruton
 - **Secondary:** Stanford Pupper Project
 - *Q1: In a quadruped trot gait, which diagonal legs must be in stance phase to keep the CoM stable?*

11. Control Systems & Debugging

- **Control Systems:** Open Loop, Closed Loop, PID Basics.
 - **Primary:** Brian Douglas Control Systems
 - **Secondary:** Control Systems Lectures (Steve Brunton)
- **Advanced Hardware Debugging (Oscilloscope):** Waveforms, Pulse Width.
 - *Q1: How would you use an oscilloscope to prove a servo is receiving a 1.5 ms PWM signal?*

v0.3 Walking Release & Verification

Functional & Stress Tests:

- Robot walks 2 meters forward without falling.
- **Repeatability Test:** Executes 10 consecutive flawless runs.

Performance Metrics:

- Walking speed (cm/s) and Battery runtime.

Deliverables: v0.3 Git Tag, Video proof, Project Report pushed.

PHASE 4: Sensors & Autonomy (2–3 Weeks)

12. Sensors & Ultrasonics

- **Hardware:** Ultrasonic time-of-flight, IMUs.
 - **Primary:** DroneBot Workshop
 - **Secondary:** Random Nerd Tutorials

13. Signal Processing

- **Filtering:** Moving Average, Median Filter, Hysteresis.
 - **Primary:** Phil's Lab
 - **Secondary:** DigiKey Tutorials
 - *Q1: Why is hysteresis required to prevent infinite stuttering on the exact edge of a threshold?*

14. Finite State Machines (FSM)

- **State Control:** IDLE → WALK → OBSTACLE_STOP.
 - **Primary:** Embedded Artistry
 - **Secondary:** State Machine Design Patterns
 - *Q1: How do you transition cleanly from walking momentum to a safe stop without falling forward?*

v0.4 Obstacle Detection Release & Verification

Functional Tests:

- Reliable obstacle detection and smooth deceleration.

Performance Metrics:

- Detection distance accuracy and Stop latency (milliseconds).

Deliverables: v0.4 Git Tag, Video proof, Project Report pushed.

PHASE 5: ROS 2 & Middleware

15. ROS 2 Fundamentals

- **Core Concepts:** Nodes, Topics, Services, Actions.
 - **Primary:** Articulated Robotics ROS 2 Series
 - **Secondary:** Official ROS 2 Documentation
- **micro-ROS:**
 - **Primary:** micro-ROS Official Documentation

PHASE 6: Advanced Robotics

16. Edge Autonomy

- **Inverse Kinematics (IK):** Dynamic (X, Y, Z) foot placement.
 - **Primary:** Modern Robotics
 - **Secondary:** Robotics Back-End
- **Machine Learning for Robotics:** Classification, Feature Engineering, Terrain Recognition.
 - **Primary:** Andrew Ng Machine Learning
 - **Secondary:** Hands-On ML (Aurélien Géron)
- **Path Planning:** A^* , Dijkstra, Occupancy Grids.
 - **Primary:** Sebastian Thrun Robotics Course
 - **Secondary:** Robotics Back-End